Class 11 and 12 Trigonometry Formula Sheet

https://gemini.google.com/app/6d85d78d2961b309

User prompt: make a formula sheet of trigonometric functions and equations and cover each and every formula that are used in class 11th and 12th

Response: Here is a comprehensive formula sheet covering all trigonometric functions, identities, and equations taught in Class 11th and 12th Mathematics.

1. Fundamentals & Angle Measurement

- **Degree to Radian:** $\text{Radian} = \text{Degree} \times \frac{\pi}{180^\circ}$
- **Radian to Degree:** $\text{Degree} = \text{Radian} \times \frac{180^\circ}{\pi}$
- **Arc Length Formula:** $s = r\theta$ (θ in radians)
- **Area of a Sector:** $A = \frac{1}{2}r^2\theta$

2. Basic Trigonometric Ratios & Reciprocals

- $\sin\theta = \frac{\text{Opposite}}{\text{Hypotenuse}}, \quad \csc\theta = \frac{1}{\sin\theta}$
- $\cos\theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}, \quad \sec\theta = \frac{1}{\cos\theta}$
- $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{\text{Opposite}}{\text{Adjacent}}, \quad \cot\theta = \frac{\cos\theta}{\sin\theta} = \frac{1}{\tan\theta}$

3. Fundamental Pythagorean Identities

- $\sin^2\theta + \cos^2\theta = 1$
- $1 + \tan^2\theta = \sec^2\theta$
- $1 + \cot^2\theta = \csc^2\theta$

4. Sign of Trigonometric Functions (ASTC Rule)

- **Quadrant I (0 to $\frac{\pi}{2}$):** All functions are **Positive** (A)
- **Quadrant II ($\frac{\pi}{2}$ to π):** $\sin$ and $\csc$ are **Positive** (S)
- **Quadrant III (π to $\frac{3\pi}{2}$):** $\tan$ and $\cot$ are **Positive** (T)
- **Quadrant IV ($\frac{3\pi}{2}$ to 2π):** $\cos$ and $\sec$ are **Positive** (C)

5. Domain, Range, and Periodicity

Function	Domain	Range	Period
$\sin x$	$\mathbb{R}$	$[-1, 1]$	2π
$\cos x$	$\mathbb{R}$	$[-1, 1]$	2π
$\tan x$	$\mathbb{R} \setminus \left\{(2n+1)\frac{\pi}{2}\right\}$	$\mathbb{R}$	π
$\csc x$	$\mathbb{R} \setminus \{n\pi\}$	$(-\infty, -1] \cup [1, \infty)$	2π
$\sec x$	$\mathbb{R} \setminus \left\{(2n+1)\frac{\pi}{2}\right\}$	$(-\infty, -1] \cup [1, \infty)$	2π
$\cot x$	$\mathbb{R} \setminus \{n\pi\}$	$\mathbb{R}$	π

6. Negative Angle Identities (Parity)

- $\sin(-x) = -\sin x$
- $\cos(-x) = \cos x$
- $\tan(-x) = -\tan x$
- $\csc(-x) = -\csc x$
- $\sec(-x) = \sec x$
- $\cot(-x) = -\cot x$

7. Allied Angle Formulas

Complementary Angles ($\frac{\pi}{2} \pm x$)

- $\sin\left(\frac{\pi}{2} - x\right) = \cos x, \quad \sin\left(\frac{\pi}{2} + x\right) = \cos x$
- $\cos\left(\frac{\pi}{2} - x\right) = \sin x, \quad \cos\left(\frac{\pi}{2} + x\right) = -\sin x$
- $\tan\left(\frac{\pi}{2} - x\right) = \cot x, \quad \tan\left(\frac{\pi}{2} + x\right) = -\cot x$

Supplementary Angles ($\pi \pm x$)

- $\sin(\pi - x) = \sin x, \quad \sin(\pi + x) = -\sin x$
- $\cos(\pi - x) = -\cos x, \quad \cos(\pi + x) = -\cos x$

- $\tan(\pi - x) = -\tan x, \quad \tan(\pi + x) = \tan x$

Multiples of 2π ($2\pi \pm x$)

- $\sin(2\pi - x) = -\sin x, \quad \sin(2\pi + x) = \sin x$
- $\cos(2\pi - x) = \cos x, \quad \cos(2\pi + x) = \cos x$
- $\tan(2\pi - x) = -\tan x, \quad \tan(2\pi + x) = \tan x$

8. Compound Angle Formulas (Addition & Subtraction)

- $\sin(A + B) = \sin A \cos B + \cos A \sin B$
- $\sin(A - B) = \sin A \cos B - \cos A \sin B$
- $\cos(A + B) = \cos A \cos B - \sin A \sin B$
- $\cos(A - B) = \cos A \cos B + \sin A \sin B$
- $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$
- $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$
- $\cot(A + B) = \frac{\cot A \cot B - 1}{\cot B + \cot A}$
- $\cot(A - B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$
- $\sin(A + B)\sin(A - B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$
- $\cos(A + B)\cos(A - B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$

9. Transformation Formulas

Product to Sum/Difference

- $2 \sin A \cos B = \sin(A + B) + \sin(A - B)$
- $2 \cos A \sin B = \sin(A + B) - \sin(A - B)$
- $2 \cos A \cos B = \cos(A + B) + \cos(A - B)$
- $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$

Sum/Difference to Product (C and D Formulas)

- $\sin C + \sin D = 2 \sin\left(\frac{C+D}{2}\right) \cos\left(\frac{C-D}{2}\right)$

- $\sin C - \sin D = 2\cos\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$
- $\cos C + \cos D = 2\cos\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$
- $\cos C - \cos D = -2\sin\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right) = 2\sin\left(\frac{C+D}{2}\right)\sin\left(\frac{D-C}{2}\right)$

10. Multiple & Sub-Multiple Angle Formulas

Double Angle ($2x$)

- $\sin 2x = 2\sin x\cos x = \frac{2\tan x}{1+\tan^2 x}$
- $\cos 2x = \cos^2 x - \sin^2 x = 2\cos^2 x - 1 = 1 - 2\sin^2 x = \frac{1-\tan^2 x}{1+\tan^2 x}$
- $1 + \cos 2x = 2\cos^2 x \implies \cos x = \pm \sqrt{\frac{1+\cos 2x}{2}}$
- $1 - \cos 2x = 2\sin^2 x \implies \sin x = \pm \sqrt{\frac{1-\cos 2x}{2}}$
- $\tan 2x = \frac{2\tan x}{1-\tan^2 x}$

Triple Angle ($3x$)

- $\sin 3x = 3\sin x - 4\sin^3 x$
- $\cos 3x = 4\cos^3 x - 3\cos x$
- $\tan 3x = \frac{3\tan x - \tan^3 x}{1-3\tan^2 x}$

11. Trigonometric Equations (General Solutions)

- $\sin x = 0 \implies x = n\pi, \quad n \in \mathbb{Z}$
- $\cos x = 0 \implies x = (2n+1)\frac{\pi}{2}, \quad n \in \mathbb{Z}$
- $\tan x = 0 \implies x = n\pi, \quad n \in \mathbb{Z}$
- $\sin x = \sin\alpha \implies x = n\pi + (-1)^n\alpha, \quad n \in \mathbb{Z}$
- $\cos x = \cos\alpha \implies x = 2n\pi \pm \alpha, \quad n \in \mathbb{Z}$
- $\tan x = \tan\alpha \implies x = n\pi + \alpha, \quad n \in \mathbb{Z}$
- $\sin^2 x = \sin^2\alpha$ or $\cos^2 x = \cos^2\alpha$ or $\tan^2 x = \tan^2\alpha \implies x = n\pi \pm \alpha, \quad n \in \mathbb{Z}$

12. Inverse Trigonometric Functions (Class 12)

Principal Value Branches

Function	Domain	Principal Value Branch (Range)
$\sin^{-1} x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1} x$	$\mathbb{R}$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
$\csc^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \setminus \{0\}$
$\sec^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$[0, \pi] \setminus \left\{\frac{\pi}{2}\right\}$
$\cot^{-1} x$	$\mathbb{R}$	$(0, \pi)$

Properties of Inverse Functions

- $\sin(\sin^{-1} x) = x \quad \text{for } x \in [-1, 1]$
- $\sin^{-1}(\sin x) = x \quad \text{for } x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Negative Arguments

- $\sin^{-1}(-x) = -\sin^{-1} x$
- $\csc^{-1}(-x) = -\csc^{-1} x$
- $\tan^{-1}(-x) = -\tan^{-1} x$
- $\cos^{-1}(-x) = \pi - \cos^{-1} x$
- $\sec^{-1}(-x) = \pi - \sec^{-1} x$
- $\cot^{-1}(-x) = \pi - \cot^{-1} x$

Reciprocal Relations

- $\csc^{-1} x = \sin^{-1}\left(\frac{1}{x}\right), \quad x \geq 1 \text{ or } x \leq -1$
- $\sec^{-1} x = \cos^{-1}\left(\frac{1}{x}\right), \quad x \geq 1 \text{ or } x \leq -1$
- $\cot^{-1} x = \begin{cases} \tan^{-1}\left(\frac{1}{x}\right), & x > 0 \\ \pi + \tan^{-1}\left(\frac{1}{x}\right), & x < 0 \end{cases}$

Complementary Sums

- $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}, \quad x \in [-1, 1]$
- $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}, \quad x \in \mathbb{R}$
- $\sec^{-1} x + \csc^{-1} x = \frac{\pi}{2}, \quad |x| \geq 1$

Sum & Difference Formulas

- $\tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x+y}{1-xy}\right), \quad xy < 1$
- $\tan^{-1} x - \tan^{-1} y = \tan^{-1}\left(\frac{x-y}{1+xy}\right), \quad xy > -1$

Multiple Angle Formulas

- $2\tan^{-1} x = \sin^{-1}\left(\frac{2x}{1+x^2}\right) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \tan^{-1}\left(\frac{2x}{1-x^2}\right) \quad (|x| < 1 \text{ for } \tan)$

13. Trigonometric Differentiation & Integration (Class 12 Calculus)

Derivatives

- $\frac{d}{dx}(\sin x) = \cos x$
- $\frac{d}{dx}(\cos x) = -\sin x$
- $\frac{d}{dx}(\tan x) = \sec^2 x$
- $\frac{d}{dx}(\sec x) = \sec x \tan x$
- $\frac{d}{dx}(\csc x) = -\csc x \cot x$
- $\frac{d}{dx}(\cot x) = -\csc^2 x$
- $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{1-x^2}$
- $\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{1-x^2}$
- $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$

Standard Integrals

- $\int \sin x \, dx = -\cos x + C$
- $\int \cos x \, dx = \sin x + C$

- $\int \sec^2 x \, dx = \tan x + C$
- $\int \csc^2 x \, dx = -\cot x + C$
- $\int \sec x \tan x \, dx = \sec x + C$
- $\int \csc x \cot x \, dx = -\csc x + C$
- $\int \tan x \, dx = \ln|\sec x| + C = -\ln|\cos x| + C$
- $\int \cot x \, dx = \ln|\sin x| + C$
- $\int \sec x \, dx = \ln|\sec x + \tan x| + C = \ln\left|\tan\left(\frac{x}{2} + \frac{\pi}{4}\right)\right| + C$
- $\int \csc x \, dx = \ln|\csc x - \cot x| + C = \ln\left|\tan\left(\frac{x}{2}\right)\right| + C$

14. Useful Maxima / Minima Expressions

- For $a\sin x + b\cos x$:
  - **Maximum value:** $\sqrt{a^2 + b^2}$
  - **Minimum value:** $-\sqrt{a^2 + b^2}$
